

HDOC DAY-37

Title: String Manipulation: Uppercase Conversion, Reversal, and Palindrome Validation

1. Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Convert a lowercase string to uppercase without using built-in functions.
2. Reverse a string.
3. Check if a string is a palindrome.
4. Exit.

2. Theoretical Background & Logic

1. ASCII Case Conversion:

In ASCII encoding, lower-case alphabets ('a' to 'z') span from values 97 to 122, while upper-case alphabets ('A' to 'Z') span from 65 to 90. Subtracting 32 from any lowercase character converts it to its corresponding uppercase character ($ch - 32$).

2. String Reversal:

The i -th character from the end of the input string is mapped to the i -th position of the output buffer: `output[i] = input[length - 1 - i]`.

3. Palindrome Checking:

A two-pointer strategy (one pointer at index 0 and one at `length - 1`) compares characters inward toward the center. If any pair of characters fails to match (`str[start] != str[end]`), the string is classified as non-palindromic.

3. Test Case Table

Test Case	Selected Option	Input String	Expected Output	Actual Output	Status
1	Lowercase to Uppercase	"upes"	"UPES"	"UPES"	Pass
2	Reverse String	"hello"	"olleh"	"olleh"	Pass
3	Check Palindrome	"racecar"	Is a palindrome	Is a palindrome	Pass

Tes t Ca se	Selecte d Option	Input String	Expect ed Output	Actual Output	Stat us
4	Check Palindr ome	"compu ter"	Is NOT a palindr ome	Is NOT a palindr ome	Pass
5	Exit	Option 5	Termina te progra m	Termina ted progra m	Pass

4. Source Code Listing

```
#include <stdio.h>

int get_length(const char str[]) {
    int len = 0;
    while (str[len] != '\0') {
        len++;
    }
    return len;
}

void to_uppercase(const char input[], char output[]) {
    int i = 0;
    while (input[i] != '\0') {
        if (input[i] >= 'a' && input[i] <= 'z') {
            output[i] = input[i] - 32;
        } else {
```

```

        output[i] = input[i];
    }
    i++;
}
output[i] = '\0';
}

void reverse_string(const char input[], char output[]) {
    int len = get_length(input);
    for (int i = 0; i < len; i++) {
        output[i] = input[len - 1 - i];
    }
    output[len] = '\0';
}

int is_palindrome(const char str[]) {
    int start = 0;
    int end = get_length(str) - 1;

    while (start < end) {
        if (str[start] != str[end]) {
            return 0;
        }
        start++;
        end--;
    }
    return 1;
}

void read_string(char str[]) {
    printf("Enter a string: ");
    scanf("%[^\\n]", str);
}

int main(void) {
    char str[200];
    char result[200];
    int choice;

    read_string(str);

```

```

do {
    printf("\n===== STRING OPERATIONS MENU
=====\\n");
    printf("1. Convert to Uppercase (without built-in functions)\\n");
    printf("2. Reverse String\\n");
    printf("3. Check if Palindrome\\n");
    printf("4. Enter a new string\\n");
    printf("5. Exit\\n");
    printf("Enter your choice (1-5): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            to_uppercase(str, result);
            printf("\\nOriginal String : %s\\n", str);
            printf("Uppercase String : %s\\n", result);
            break;

        case 2:
            reverse_string(str, result);
            printf("\\nOriginal String : %s\\n", str);
            printf("Reversed String : %s\\n", result);
            break;

        case 3:
            if (is_palindrome(str)) {
                printf("\\n\\\"%s\\\" IS a palindrome.\\n", str);
            } else {
                printf("\\n\\\"%s\\\" is NOT a palindrome.\\n", str);
            }
            break;

        case 4:
            read_string(str);
            break;

        case 5:
            printf("Exiting program...\\n");
            break;
    }
} while (choice != 5);

```

```

        default:
            printf("Error: Invalid choice! Please select between 1 and 5.\n");
            break;
    }
} while (choice != 5);

return 0;
}

```

4. Execution Screenshots

```

kali@kali:~/HDOOC_Day37
$ mkdir HDOOC_Day37
$ cd HDOOC_Day37
$ nano day37.c
$ gcc day37.c -o day37
$ ./day37
Enter a string: madam

===== STRING OPERATIONS MENU =====
1. Convert to Uppercase (without built-in functions)
2. Reverse String
3. Check if Palindrome
4. Enter a new string
5. Exit
Enter your choice (1-5): 1

Original String : madam
Uppercase String : MADAM

===== STRING OPERATIONS MENU =====
1. Convert to Uppercase (without built-in functions)
2. Reverse String
3. Check if Palindrome
4. Enter a new string
5. Exit
Enter your choice (1-5): 2

Original String : madam
Reversed String : madam

===== STRING OPERATIONS MENU =====
1. Convert to Uppercase (without built-in functions)
2. Reverse String
3. Check if Palindrome
4. Enter a new string
5. Exit
Enter your choice (1-5): 3

"madam" IS a palindrome.

===== STRING OPERATIONS MENU =====
1. Convert to Uppercase (without built-in functions)
2. Reverse String
3. Check if Palindrome
4. Enter a new string
5. Exit
Enter your choice (1-5):

```